

Uncertainty

Exercise 6 — ISS dataviz, Spring 2026

YOUR NAME HERE

Invalid Date

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Task 1: Session check-in

Three interesting or exciting things:

1. Something
2. Something
3. Something

Three muddy or unclear things:

1. Something
2. Something
3. Something

Task 2: US public spending on children

Data description and cleaning

This data comes from the Urban Institute, and it was a #TidyTuesday dataset in September 2020

This dataset is a little different from others you've worked with for a couple reasons!

1. The data has a column for each state, but does not come with columns for US Census regions (i.e. South, West, Midwest, and Northeast) or US Census divisions (e.g., Middle Atlantic, New

England, Pacific, etc.). To give you those columns, I've included a second dataset in `data/state_details.csv` that contains these details—in the code below you'll use `left_join()` to merge this dataset with the main data.

2. The main `data/kids.csv` file is long and tidy, which means that instead of having a column for education expenditures, a column for Pell Grants, a column for Head Start expenditures, and so on, there's one column called `variable` with rows for each of the different variables, like this:

	state	variable	year	raw	inf_adj	inf_adj_perchild
	<chr>	<chr>	<dbl>	<dbl>	<dbl>	<dbl>
1	Alabama	PK12ed	1997	3271969	4665308.	3.93
2	Alabama	highered	1997	956505	1363824.	1.15
3	Alabama	edsubs	1997	107733	153610.	0.129
4	Alabama	edservs	1997	246057	350838.	0.296
5	Alabama	pell	1997	120833.	172289.	0.145
6	Alabama	HeadStartPriv	1997	55777.	79530.	0.0670
7	Alabama	TANFbasic	1997	63818.	90994.	0.0766

This means you'll need to use `filter()` to choose the different variables you want to work with:

```
k12_spending <- kids |>
  filter(variable == "PK12ed")
```

After merging the two datasets, you'll have 9 columns in `kids`:

- `state`: The state (including the District of Columbia)
- `variable`: The variable being measured. There are 23 of these—see the codebook here for a complete list. You'll be working with these in this exercise:
 - `PK12ed`: Elementary and secondary education expenditures
 - `SNAP`: Public spending on SNAP benefit payments that go to children
 - `lib`: Public spending on libraries
- `year`: The year
- `raw`: The actual amount spent
- `inf_adj`: The amount spent, adjusted to 2016 dollars
- `inf_adj_perchild`: The amount spent *per child in the state*, adjusted to 2016 dollars
- `state_abb`: The state abbreviation (e.g., AL, AK, AZ)
- `region`: The US Census region (South, West, Midwest, and Northeast)
- `division`: The US Census division (e.g., Middle Atlantic, New England, Pacific)

The code below loads, merges, and cleans the two datasets, resulting in a data frame named `kids`. **You don't need to modify anything in this chunk of code—you only need to run it.**

```

library(tidyverse) # For ggplot2, dplyr, and friends
library(ggthemes)  # For ridge plots
library(ggthemes)  # For half geoms

# Make the randomness of jittering points the same every time you render this
set.seed(1234)

# Load state data
state_details <- read_csv("data/state_details.csv")

# Load kids data and merge in state data
kids <- read_csv("data/kids.csv") |>
  left_join(state_details, by = join_by(state)) |>
  # These values are stored as 1000s of dollars, so this scales them back up
  by
  # multiplying them by 1000
  mutate(across(c(raw, inf_adj, inf_adj_perchild), \(x) x * 1000))

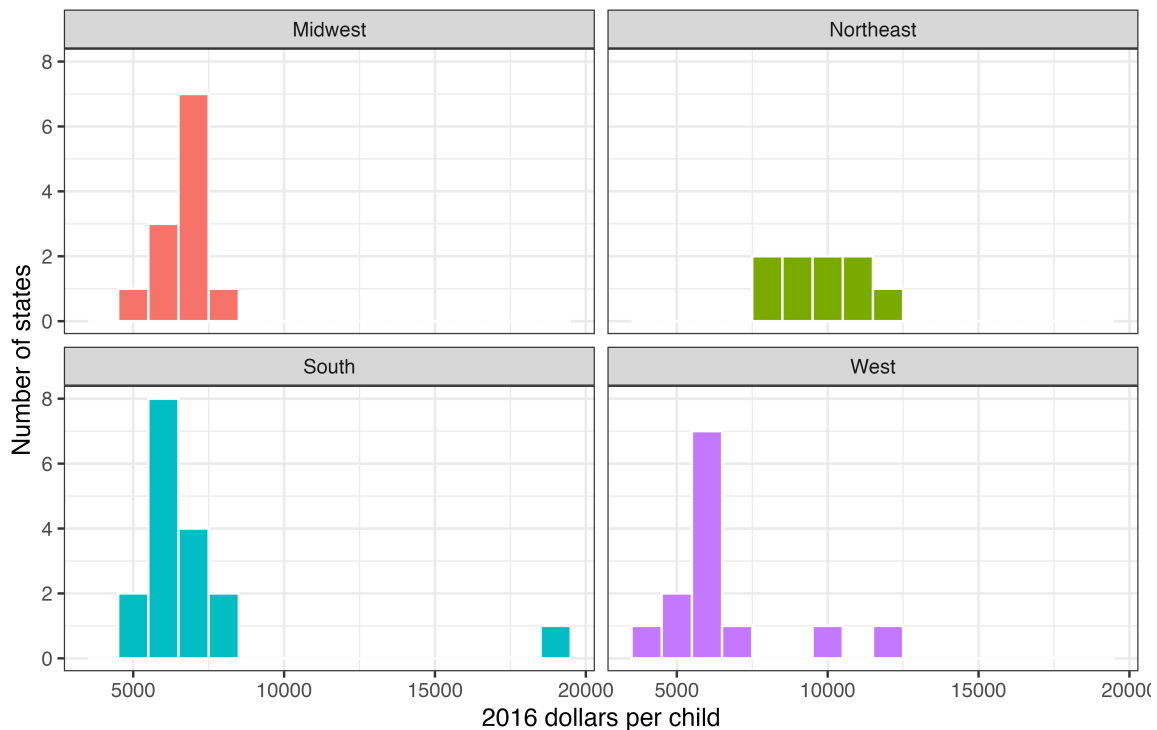
```

Histograms

Run the code chunk below to see a plot. It shows the distribution of **inflation-adjusted, per child spending on K-12 education in 2010** across the four US Census regions.

Spending on K-12 education per child in 2010

Amounts adjusted for inflation to 2016 dollars



Use `ggplot()` and `geom_histogram()` in the chunk below to re-create the plot above using the `kids` dataset. **Some hints:**

- You'll need to make a smaller filtered dataset first and then plot *that*, just like you did with exercise 3 and 4.
 - You'll need to filter the data so it only includes rows for elementary and secondary education spending (`variable == "PK12ed"`)
 - You'll need to filter the data so it only shows 2010 data
- You'll need to facet by region and remove the legend
- You'll need to choose a good binwidth (the image above uses \$1,000 per bin)
- The image above uses `theme_bw()`

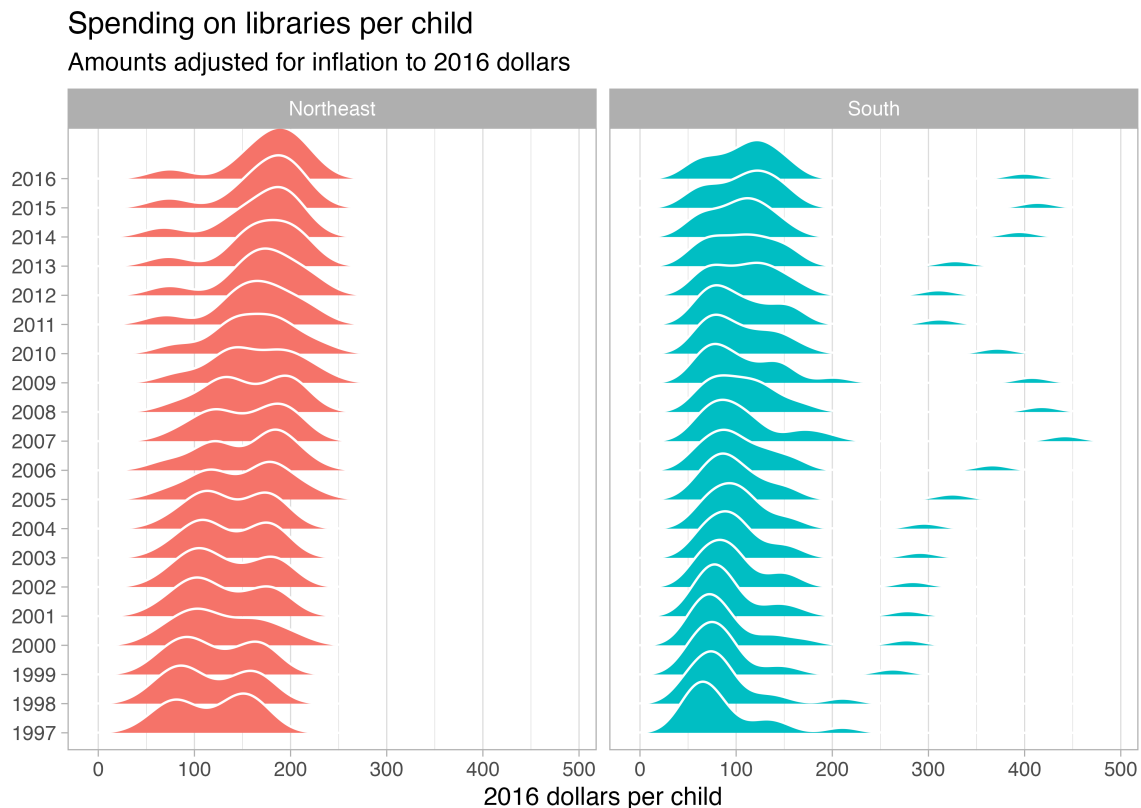
```
# Do stuff here
```

Interpret this plot in a few sentences. What does this show?

DO THAT HERE

Densities

Run the code chunk below to see a plot. It shows the distribution of **inflation-adjusted, per child spending on libraries in the Northeast and the South** from 1997 to 2016.



Use `ggplot()` and `geom_density_ridges()` in the chunk below to re-create the plot above using the `kids` dataset. **Some hints:**

- You'll need to make a smaller filtered dataset first and then plot *that*, just like you did with exercise 3 and 4.
 - You'll need to filter the data so it only includes rows for library spending (`variable == "lib"`)
 - You'll need to filter the data so it only shows data for the south and northeast (`region %in% c("South", "Northeast")`)
- You'll need to facet by region and remove the legend
- If you use `y = year`, you'll get an error. `geom_density_ridges()` needs a categorical variable on the y-axis to work. You can convert `year` to categorical by making it a *factor* (what R calls categorical variables). You can do that inside `aes()`, like `aes(..., y = factor(year))`.
- Add a white border to `geom_density_ridges()`
- The image above uses `theme_light()` and it uses `theme()` to disable the major y-axis gridlines (`panel.grid.major.y = element_blank()`)

```
# Do stuff here
```

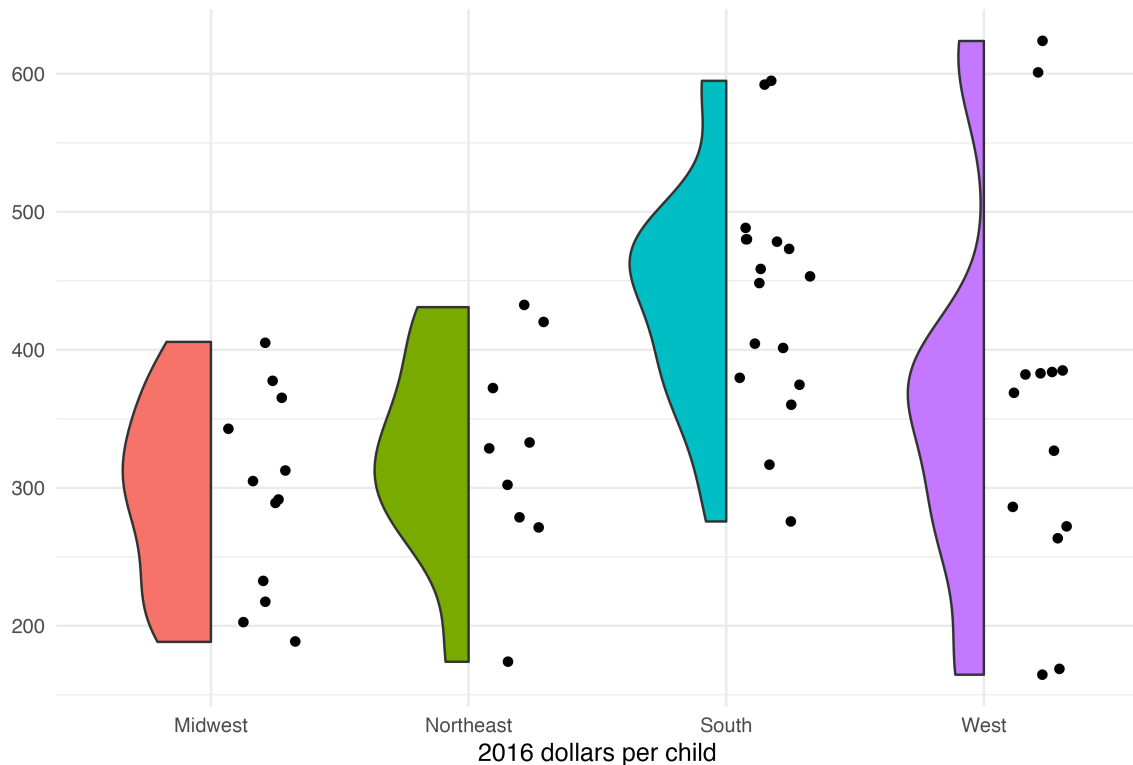
Interpret this plot in a few sentences. What does this show?

DO THAT HERE

Boxes, violins, and friends

Run the code chunk below to see a plot. It shows the distribution of **inflation-adjusted, per child spending on SNAP** in **2016** across all four US census regions.

Spending on SNAP per child in 2016



Use `ggplot()` and `geom_half_point()` and `geom_half_violin()` in the chunk below to re-create the plot above using the `kids` dataset. **Some hints:**

- You'll need to make a smaller filtered dataset first and then plot *that*, just like you did with exercise 3 and 4.
 - You'll need to filter the data so it only includes rows for SNAP spending (`variable == "SNAP"`)
 - You'll need to filter the data so it only shows data for 2016
- You'll need to facet by region and remove the legend
- The image above uses `theme_minimal()`

```
# Do stuff here
```

Interpret this plot in a few sentences. What does this show?

DO THAT HERE

Task 3: Extension

Copy the code for one of your plots above and paste it into the chunk below. Do some extra things to it, like using a different variable (there are lots of them!), filtering for a specific region or division, looking at different years, changing the labels, changing the colors, adding a new geom, changing the theme, or whatever. **This is your chance to play around and explore.**

Do stuff here

Bibliography